Taking Attention Out of Context: Windowed RoPE, NoPE Retrieval, and Squared Logits for 32× Length Generalization

Lucina
ritualistic.invocation@gmail.com
github.com/6elphegor/kvbench

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Abstract

Transformers trained at one context length typically fail to retrieve in-context information from much longer contexts. We isolate this failure with a minimal synthetic benchmark: sequences of shuffled key–value entries in which every key appears exactly twice with the same value, and the model must reproduce the value at the key's second occurrence: a pure in-context lookup with no linguistic confounds. We train eight ~2.1M-parameter decoder-only transformers that share an identical backbone and differ only in their attention layers, along two axes: positional treatment (full RoPE, partial RoPE, or interleaved local-RoPE / global-NoPE, with and without a sliding window on the RoPE layers) and logit sharpening (standard softmax(s) versus squared logits, softmax(s^2)). Trained on contexts of 8 keys (128 tokens), only the interleaved variants with a sliding window generalize: with squared logits, exact-match retrieval holds at ~99.9% out to 256 keys (4,096 tokens, 32× the training length), while full-RoPE, partial-RoPE, and window-free interleaved variants all collapse toward zero. The sliding window on RoPE layers is necessary, not incidental: without it the interleaved variant fails like plain RoPE. A ninth variant replaces the windowed-RoPE layers with Kimi Delta Attention (KDA), the gated delta-rule linear attention used in Kimi K3: it is the only variant without a sliding window that avoids collapse, yet its learned fading memory degrades smoothly to ~75% at 32×, suggesting that the window's *exact* locality, not locality per se, underwrites flat generalization. A simple variance analysis explains the exponent: for Gaussian logits, $p = 2$ is the unique power for which e^{s^p} has a regularly varying tail, so softmax weights neither diffuse (as for $p < 2$) nor collapse onto a single token (as for $p > 2$) as context grows. We also document two optimization effects: a short context-length curriculum is required to keep full-attention RoPE variants off a positional-shortcut local optimum, and the plain interleaved variant is seed-sensitive. Code, specs, and figures are public.

1 Introduction

Length generalization (performing at test time on longer inputs than seen in training) remains a persistent failure mode of transformers [1, 20]. For modern LLMs the practically important instance is *retrieval*: can the model find and reproduce information stored earlier in a context longer than any it was trained on? Long-context benchmarks such as needle-in-a-haystack [6] and RULER [4] show that even models trained at long context often degrade well before their nominal context limit; models evaluated *beyond* their training length usually fail outright.

Two mechanisms plausibly drive this failure. First, *positional*: rotary position embeddings [16] place unseen relative distances out of distribution, and a large literature exists on patching this post hoc via interpolation and frequency scaling [2, 13]. Second, *dispersal*: with more tokens competing in each softmax, attention mass spreads and the retrieved value signal is diluted, an effect analyzed by Veličković et al. [17] and addressed by length-aware sharpening such as Scalable-Softmax [11].

This paper studies both mechanisms jointly in a setting small and clean enough to give an unambiguous answer. We construct a synthetic key–value retrieval task (Section 2) and train a grid of eight architecture variants that share an identical ~2.1M-parameter backbone and differ *only* in their attention layers (Section 3): the positional axis crosses full RoPE, partial RoPE, and interleaved local-RoPE / global-NoPE layers (with and without a sliding window on the RoPE layers); the sharpening axis toggles squaring of attention logits before the softmax. A ninth variant, described in Section 3, replaces the windowed-RoPE layers with Kimi Delta Attention to test whether a *learned* fading local memory can substitute for the window's structural one.

Findings.

1. **Only interleaved local-RoPE / global-NoPE attention with a sliding window generalizes.** Trained at 8 keys (128 tokens), the squared-logit interleaved variant retrieves at ~99.9% exact match out to 256 keys (4,096 tokens, 32× training length); the plain interleaved variant declines gracefully to ~94%. Full-RoPE and partial-RoPE variants collapse toward zero, *and so does the interleaved variant without the sliding window*. Confining RoPE layers to a short local window is essential, not cosmetic (Section 6).

2. **Squaring attention logits stabilizes retrieval against context growth, and the exponent 2 is critical.** If logits are approximately Gaussian, the weight e^{s^p} has a regularly varying (power-law) tail exactly at $p = 2$: for $p < 2$ softmax weights obey a law of large numbers and diffuse as context grows; for $p > 2$ the maximum dominates and attention collapses onto one token; at $p = 2$ the weight distribution neither diffuses nor degenerates (Section 4). Empirically, squared logits turn the interleaved variant's slow decay into flat ~99.9% retrieval, but do not rescue any variant whose positional scheme is broken.

3. **A learned fading memory (KDA) avoids collapse but not decay.** Replacing the windowed-RoPE layers of the winning variant with Kimi Delta Attention [8, 9] yields the only non-windowed variant that does not collapse: retrieval is perfect to 3× the training length and still ~75% at 32×, but it does not match the window it replaced. Exact, structural locality appears to be what converts graceful degradation into flat generalization (Section 6).

4. **Optimization details decide which basin is reached.** Trained directly at 8 keys, every variant whose RoPE layers see full attention lands on a positional-shortcut local optimum and plateaus at ~37% retrieval *in distribution*; a short context-length curriculum ($n = 2 \to 4 \to 6 \to 8$ keys over the first 2,500 of 15,000 steps) avoids this entirely. The plain interleaved variant is additionally seed-sensitive (Section 5).

Our contribution is deliberately narrow-and-controlled rather than broad-and-confounded: a factorial comparison in which the *only* difference between variants is the attention layer, on a task where retrieval is the entire objective. The result is a clean existence proof and failure catalogue at small scale, consistent with, and mechanistically clarifying, the design trend toward interleaved local/global attention with position-free global layers in recent production models [3, 10].

2 The task: paired key–value retrieval

The vocabulary has 12 tokens: digits `0--9`, `StartKey`, `StartValue`. An *entry* is `StartKey` $k_1 k_2 k_3$ `StartValue` $v_1 v_2 v_3$: 8 tokens with 3-digit keys and values. A sequence with n unique keys contains $2n$ entries: every key appears *exactly twice* with the same (independently random) value, and the $2n$ entries are shuffled uniformly. Sequences are generated on the fly; the training distribution is effectively inexhaustible.

The first occurrence of a key carries an unpredictable value; the second occurrence repeats it, so it is exactly recoverable by in-context lookup. The training loss is cross-entropy on the value digits of *second-occurrence* entries only; first-occurrence values and all structural positions are excluded, so every gradient term corresponds to a solvable retrieval. The evaluation metric is *exact-match retrieval accuracy*: the fraction of second occurrences whose value digits are all predicted correctly.

Training uses $n = 8$ keys (16 entries, 128 tokens). Evaluation sweeps $n \in \{8, 12, 16, 24, 32, 48, 64, 96, 128, 160, 192, 256\}$; at $n = 256$ the context is 4,096 tokens, 32× the training length.

This task is a distilled form of the key–value retrieval subtasks in long-context suites [4, 6], with two useful properties: chance performance is 10^{-3} (guessing 3 digits), so success is unambiguous; and the model must *copy from context* rather than memorize, since values are random per sequence. Mechanistically, the minimal solution is an induction-style circuit [12] keyed on 3-token key matches.

3 Nine attention variants, one backbone

All variants share a standard decoder-only backbone: token embedding, 8 transformer blocks, final LayerNorm, and output projection. Each block is x += Attn(LN(x)) followed by x += FFN(LN(x)), with learnable bias-free LayerNorm, no biases anywhere, SwiGLU FFNs (hidden $4d$), multi-head attention with 4 query and 4 KV heads, $d_{\text{model}} = 128$,

variant	attention layers	squared logits
`rope` / `rope_square`	full RoPE, full attention, every layer	no / yes
`partial_rope` / `partial_rope_square`	half-RoPE, full attention, every layer	no / yes
`hybrid` / `hybrid_square`	alternating [RoPE, window W=10] and [NoPE, full]	no / yes
`hybrid_nowindow` / `hybrid_square_nowindow`	alternating [RoPE, full] and [NoPE, full]	no / yes
`hybrid_square_kda`	alternating [KDA] and [NoPE, full]	NoPE layers

Table 1: The variant grid. "Alternating" interleaves the two attention types across the 8 attention layers (4 of each). All variants share the identical backbone and training recipe. The ninth variant replaces the windowed-RoPE layers of `hybrid_square` with Kimi Delta Attention; KDA has no softmax, so squared logits apply only to its NoPE layers.

head dimension 32, standard $1/\sqrt{d_k}$ scaling, causal masking, and default PyTorch initialization, for ~2.1M parameters in total. Variants differ only inside the attention layers, along three axes:

- **RoPE fraction**: rotary embeddings [16] applied to all of each head's dimensions (full RoPE), half (partial RoPE, as in several production models), or none (NoPE [7]).
- **Sliding window**: local causal attention of width $W = 10$ tokens (slightly more than one 8-token entry), or full attention.
- **Squared logits**: $\mathrm{softmax}(s^2)$ in place of $\mathrm{softmax}(s)$, applied elementwise to the scaled scores before masking.

The eight variants (Table 1) populate this grid. The interleaved ("hybrid") pattern alternates a *local* layer (full RoPE, sliding window $W = 10$) with a *global* layer (no positional encoding, full attention). The intent: the NoPE layers perform purely content-based lookup, which is length-agnostic by construction, while RoPE is confined to relative distances $< W$ that never leave the training distribution no matter how long the context grows. The `nowindow` ablation removes only the sliding window, testing whether interleaving NoPE layers alone suffices.

A ninth variant: Kimi Delta Attention in place of the window. Motivated by production hybrids that interleave linear-attention layers with periodic position-free global attention [8, 9], `hybrid_square_kda` keeps the winning `hybrid_square` layout but replaces each windowed-RoPE layer with Kimi Delta Attention (KDA): a delta-rule fast weight memory [15, 19] with a *channel-wise* forget gate. Each head maintains a recurrent state $S_t \in \mathbb{R}^{d_k \times d_v}$ updated as

$$S_t = \left(I - \beta_t k_t k_t^\top\right) \mathrm{Diag}(a_t)\, S_{t-1} + \beta_t k_t v_t^\top, \qquad o_t = S_t^\top q_t, \tag{1}$$

with per-channel retention $a_t \in (0,1)^{d_k}$, write strength $\beta_t = \sigma(w_\beta^\top x_t)$, a short causal convolution and SiLU on $q/k/v$, ℓ_2-normalized q and k, and a per-head RMSNorm with sigmoid output gate. We follow the Kimi K3 parameterization [9] exactly, including its two changes over Kimi Linear: the log-decay is lower-bounded by a scaled sigmoid, $g_t = g_{\min}\, \sigma(e^{A_h} z_t)$ with $g_{\min} = -5$, and the output gate is full-rank. The layer is computed with the chunkwise WY-form parallel scan of Kimi Team [8], verified against the naive per-token recurrence to $\sim 10^{-12}$ in double precision. Like the sliding window, KDA is a fading local memory with no length-dependent parameters (no positional encoding, no attention mask); unlike the window, its locality is *learned and approximate* rather than structural and exact. The global NoPE layers keep squared logits (KDA itself has no softmax to square), and the parameter count (~2.19M) matches the grid.

4 Why squared logits, and why the exponent must be 2

As context length n grows, softmax attention becomes increasingly diffuse: with more tokens competing, the weight on the correct token shrinks and the retrieved value is diluted by averaging [17]. A minimal model captures this. Let logits $s_1, \ldots, s_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ (the scale attention's $1/\sqrt{d_k}$ factor is designed to produce), let $w = \mathrm{softmax}(s^p)$ for an elementwise power p, and let values $z_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ be independent of the logits. The variance of the attended output $\sum_i w_i z_i$ equals $\mathbb{E}\,\|w\|_2^2$, the expected inverse participation ratio of the attention distribution: it is 1 when attention concentrates on one token and $1/n$ when attention is uniform.

The tail of the weight e^{s^p} determines the regime. Since $\Pr[e^{s^p} > t] = \Pr[s > (\ln t)^{1/p}] \approx e^{-(\ln t)^{2/p}/2}$:

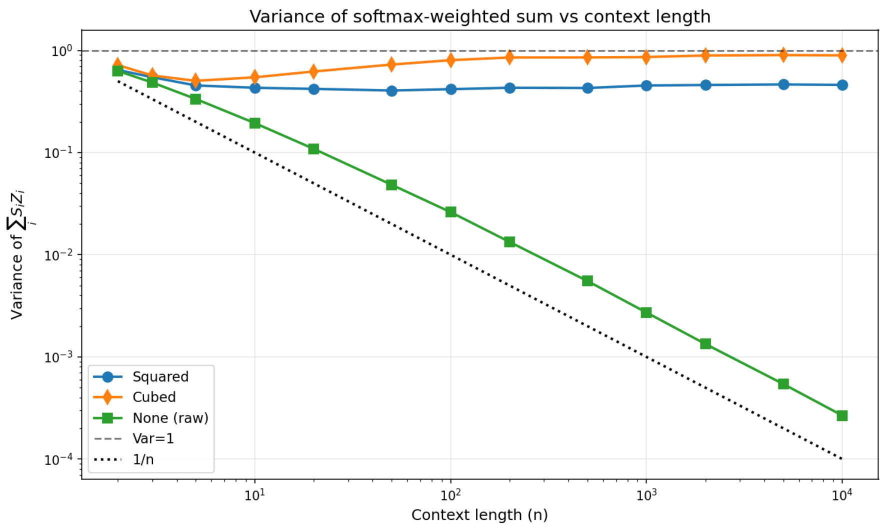

Figure 1: Monte Carlo estimate of $\mathrm{Var}[\sum_i w_i z_i]$ for $w = \mathrm{softmax}(s^p)$ with Gaussian logits and values, as a function of context length n. Standard attention ($p = 1$, "None") decays toward the $1/n$ uniform-attention floor; cubing ($p = 3$) drives the variance to 1 (single-token collapse); squaring ($p = 2$) is stable between the two for all n up to 10^4.

- $p < 2$ (including standard attention, $p = 1$): the tail decays faster than any power law, all moments are finite, and $\sum_j e^{s_j^p}$ obeys a law of large numbers. The largest weight $w_{\max} \to 0$ and $\mathbb{E}\|w\|_2^2 \to 0$: attention *diffuses* and the retrieved signal vanishes as n grows.
- $p = 2$: $\Pr[e^{s^2} > t] \sim t^{-1/2}$ up to slowly varying factors, a regularly varying tail with index $\frac{1}{2} < 1$. The sum is dominated by its few largest terms, and the normalized weights converge to a nondegenerate heavy-tailed limit: $\mathbb{E}\|w\|_2^2$ stays bounded away from both 0 and 1 for all n.
- $p > 2$: the tail is heavier than any power law; the single largest term dominates the sum, $w_{\max} \to 1$, and attention *collapses* onto one token.

Figure 1 confirms this numerically out to $n = 10^4$: $p = 1$ decays steadily toward the $1/n$ floor, $p = 3$ saturates at 1, and $p = 2$ holds a stable intermediate value across three orders of magnitude of context length. Squaring is thus not one arbitrary sharpener among many: among elementwise powers it is *the* critical exponent at which softmax attention neither washes out nor degenerates as context grows. Note also that squaring is sign-symmetric: strongly negative logits receive high weight along with strongly positive ones, so the model must (and, empirically, does) learn score distributions compatible with this.

We do not claim novelty for logit squaring itself, which we encountered in an informally circulated article; it belongs to the same family of length-robust sharpening interventions as Scalable-Softmax [11] and the temperature adjustments analyzed by Veličković et al. [17]. The experimental point here is different and, to our knowledge, new: sharpening alone rescues nothing: it converts slow decay into flat generalization *only* on top of a positional scheme that is itself length-stable (Section 6).

5 Training protocol

All variants train identically: AdamW ($\beta = (0.9, 0.999)$, weight decay 0.1), learning rate 3×10^{-4} with 500-step linear warmup and cosine decay, batch size 64, 15,000 steps, ~7 minutes per variant on a single consumer GPU. Loss is cross-entropy on second-occurrence value digits only.

Context-length curriculum. Training proceeds at $n = 2$ keys until step 750, $n = 4$ until 1,500, $n = 6$ until 2,500, then $n = 8$ thereafter. This is not a convenience but a necessity for the full-attention RoPE variants: trained at $n = 8$

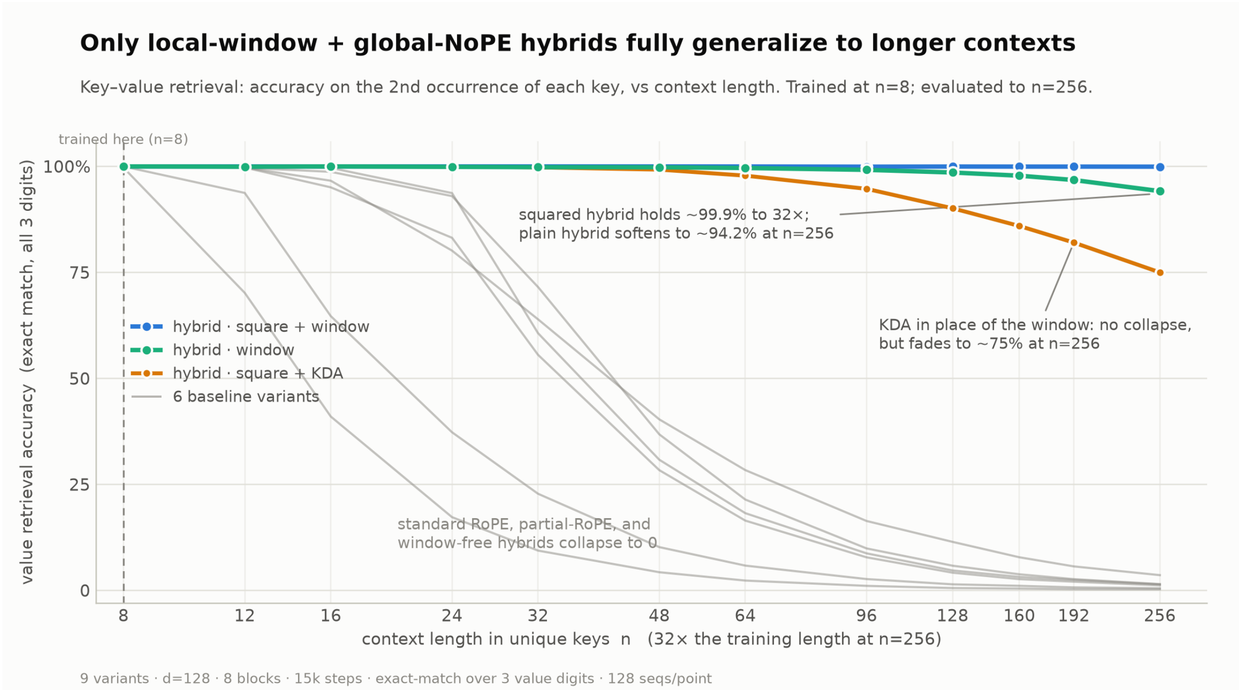

Figure 2: Exact-match retrieval accuracy (all 3 value digits) on second occurrences versus context length n (unique keys), 128 sequences per point. All variants are trained at $n = 8$ (dashed line) and reach ≈100% there. Only the windowed interleaved variants generalize: `hybrid_square` holds ∼99.9% to $n = 256$ (32× training length); plain `hybrid` declines to ∼94.2%. The KDA variant `hybrid_square_kda` avoids collapse but fades to ∼75% at $n = 256$. All six remaining variants (full RoPE, partial RoPE, and the window-free hybrids, with or without squared logits) collapse toward zero.

from the start, every variant whose RoPE layers see full attention (`rope`, `partial_rope`, both `nowindow` hybrids) falls into a local optimum that predicts values from *positional* statistics rather than content matching and plateaus at ∼37% exact match *at the training length*. Started at $n = 2$, where only 2 keys exist and no positional shortcut is available, the content-matching circuit forms first, and every variant then reaches ≈100% in-distribution accuracy. The windowed hybrids learn the task with or without the curriculum. This is consistent with reports that length generalization is sensitive to optimization trajectory rather than capacity [20].

Seeds. Every variant uses seed 0 except the plain `hybrid`, which uses seed 1: at seed 0 it converges to a partial positional shortcut and plateaus at ∼78% in-distribution retrieval. Its squared-logit counterpart shows no such sensitivity. We report this rather than hide it; single-seed results are a stated limitation (Section 8).

6 Results

In distribution, everything works. With the curriculum, all nine variants reach ≈99–100% exact-match retrieval at the training length $n = 8$. Whatever separates the variants, it is not the ability to learn the task.

Out of distribution, the factorial structure is stark (Figure 2). The two windowed interleaved variants are the only survivors. `hybrid_square` is essentially flat: ∼99.9% at $n = 256$, i.e. retrieval is unimpaired at 32× the training context. Plain `hybrid` decays gently (∼94% at $n = 256$): the diffusion effect of Section 4 at work on an otherwise sound positional scheme. Every other variant collapses: full RoPE and partial RoPE fail because unseen relative distances place their attention out of distribution [16, 14, 7], and squaring does not save them: sharpened attention to the wrong places is still wrong. Most diagnostically, `hybrid_nowindow` and `hybrid_square_nowindow` also

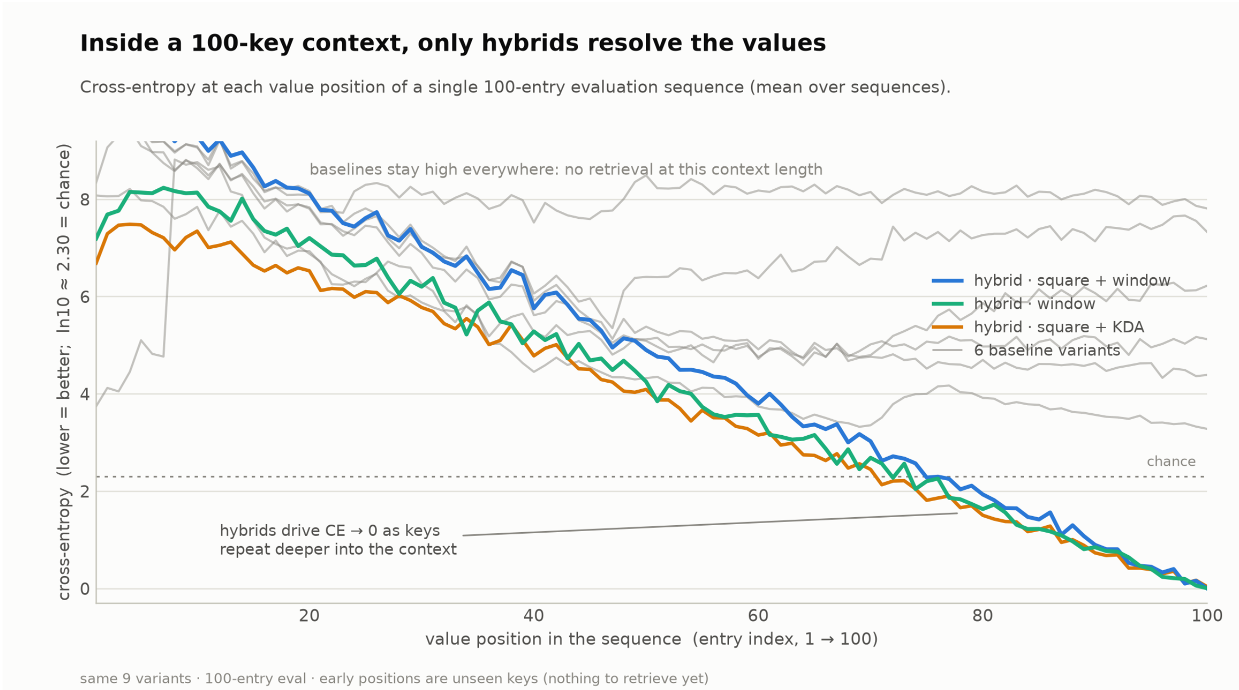

Figure 3: Mean cross-entropy per value digit at each entry index of a 100-entry evaluation sequence ($n = 50$ keys, 12.5× training length). Early entries are mostly first occurrences (nothing to retrieve), so all models sit above chance ($\ln 10 \approx 2.30$, dotted); as the sequence progresses, an increasing fraction of entries are second occurrences. Both windowed hybrids and the KDA hybrid drive CE toward 0 deep into the context; the six baselines never descend below chance at any position: they perform no retrieval at all at this length.

collapse, no better than plain RoPE: interleaving NoPE layers is not enough. If even alternate layers apply RoPE over unbounded distances, the stack inherits RoPE's failure; the sliding window is what keeps every rotary comparison inside the training distribution.

KDA: no collapse, graceful fade. The ninth variant sits in a class of its own. `hybrid_square_kda` is the only variant without a sliding window that does not collapse: retrieval is perfect through $n = 24$, ~99.8% at $n = 32$, then fades smoothly: 97.8% at $n = 64$, 90.1% at $n = 128$, 75.0% at $n = 256$. Its per-position cross-entropy at the 100-entry evaluation length is indistinguishable from the windowed hybrids' (Figure 3), so the shortfall is purely a long-context effect, not weaker retrieval. The comparison localizes what the window contributes: a $W = 10$ window computes the *identical* function at every context length, whereas KDA is only approximately local. Its decay gates are trained on 128-token sequences, and whatever state survives longer horizons than training presents is a distribution shift the downstream NoPE retrieval layers never saw; its fixed-size state must also compress an ever-longer past. The result supports a refinement of our main claim: a fading local memory interleaved with NoPE globals suffices to *avoid collapse*, but the window's exact, structural locality is what converts graceful degradation into flat generalization.

Per-position behavior (Figure 3). Within a 100-entry sequence (12.5× training length), both windowed hybrids drive cross-entropy toward zero as keys begin to repeat deeper in the context, while all six baselines remain *above* chance at every position: at this length they are not degraded retrievers but non-retrievers. The hybrids' early-position CE is above the baselines' (the cost of having no global position signal when nothing is yet retrievable), a price paid exactly where it does not matter.

7 Related work

Positional encodings and length generalization. RoPE [16] does not extrapolate; ALiBi [14] attacked train-short/test-long directly via linear distance penalties (a soft locality bias related in spirit to our hard window); interpolation and frequency-scaling methods [2, 13] patch RoPE post hoc. Kazemnejad et al. [7] showed NoPE decoders length-generalize better than explicit encodings on synthetic tasks, and Wang et al. [18] extended the case for position-free causal attention. Our contribution to this line is the controlled factorial finding that NoPE layers generalize retrieval *only* when the accompanying RoPE layers are windowed.

Interleaved local/global attention in production models. Sliding-window attention appeared in Mistral [5]; Gemma 2 [3] interleaves local and global RoPE layers; Llama 4 [10] interleaves windowed-RoPE layers with position-free global layers ("iRoPE") explicitly for long-context generalization. Our windowed hybrid is a minimal instance of this design, and our ablations isolate *which* ingredient carries the generalization (the window on rotary layers, with NoPE globals doing the lookup) at a scale where every alternative can be trained and compared exactly.

Linear attention and delta rules. Delta-rule fast weight programmers [15] replace softmax attention with a recurrent associative memory; Gated DeltaNet [19] adds a per-head forget gate, and Kimi Delta Attention [8] refines the gate to be channel-wise. Kimi Linear and Kimi K3 [9] interleave KDA with NoPE global attention (3:1 with gated MLA in K3): structurally the same recipe as our windowed hybrid, with the window replaced by a learned fading memory. Our controlled substitution (Section 6) shows the two are not equivalent under train-short/test-long: KDA preserves the non-collapse property of the recipe but trades the window's exact length-invariance for smooth degradation.

Attention dispersion and sharpening. Veličković et al. [17] prove softmax attention must disperse as input length grows and propose temperature adaptation; Scalable-Softmax [11] rescales logits with $\log n$. Elementwise squaring is a fixed, length-independent alternative sitting at the critical exponent of Section 4. Our results position sharpening as a *second-order* correction: necessary for flat (rather than slowly decaying) retrieval, sufficient for nothing on its own.

Retrieval benchmarks and mechanisms. Our task distills the key–value retrieval components of RULER [4] and needle-in-a-haystack tests [6] into a minimal, chance-calibrated form, and the underlying circuit is induction-head-like content matching [12]. Sensitivity of length generalization to optimization details echoes Zhou et al. [20] and Anil et al. [1].

8 Limitations

These results are an existence proof and failure catalogue at small scale, not a demonstration at LLM scale. Specifically: (i) models are ~2.1M parameters on a 12-token vocabulary; whether the factorial pattern survives natural language pretraining at scale is untested here, though the convergent design of production models [3, 10] is encouraging. (ii) Each variant is a single training run; we directly observed seed sensitivity in one variant (`hybrid`, seed 0 vs. 1), so per-variant accuracy numbers carry run-to-run uncertainty that we have not quantified with multi-seed error bars. (iii) The task has exact 3-token key matches; fuzzy or semantic retrieval may load differently on positional information. (iv) The variance analysis of Section 4 assumes i.i.d. Gaussian logits independent of values, a caricature of trained attention; we offer it as an explanation of the observed stability, not a theorem about trained networks. (v) Squared logits make attention sign-symmetric, and their interaction with language modeling objectives (as opposed to pure retrieval) is unexplored here. (vi) The KDA comparison inherits these caveats and adds its own: it is a single run at head dimension 32 (K3 uses 128), we did not tune $g_{\min}$, the KDA:global ratio (1:1 here versus K3's 3:1), or the training length its decay gates adapt to, and any of these could narrow the gap to the window.

9 Conclusion

On a minimal key–value retrieval task, we trained eight attention variants differing only in positional treatment and logit sharpening, and found a sharp factorial answer to context-length generalization: confine rotary position encodings to a short sliding window, let interleaved position-free layers do global content lookup, and square the attention logits; retrieval is then essentially perfect at 32× the training length, while every ablation of the positional recipe collapses to zero. The window is load-bearing, sharpening is a second-order stabilizer with a principled critical exponent, and a context-length curriculum decides whether full-attention RoPE models learn retrieval at all. A ninth variant substituting Kimi K3's delta-rule linear attention for the sliding window shows the recipe's local component can be learned rather than structural, at the price of smooth degradation (~75% at 32×) in place of flat generalization. We release the benchmark, all specs, and training code at github.com/6elphegor/kvbench; each variant trains in minutes on one consumer GPU, making this an easy substrate for testing further attention designs.

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